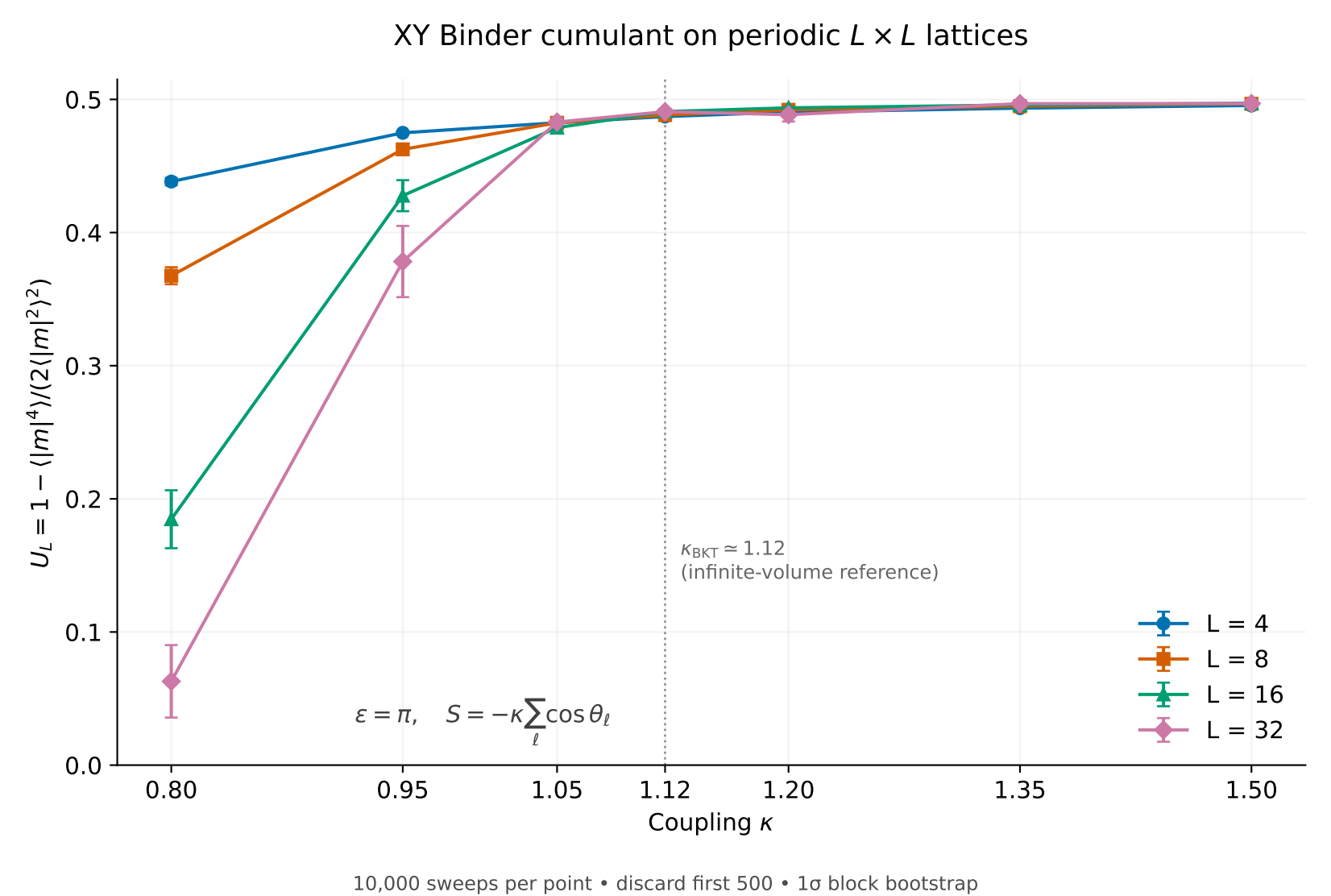
XY Binder cumulant on periodic $L \times L$ lattices
$U_L = 1 - \langle |m|^4 \rangle / (2 \langle |m|^2 \rangle^2)$
0.0
0.1
0.2
0.3
0.4
0.5
0.80
0.95
1.05
1.12
1.20
1.35
1.50
Coupling κ
$\kappa_{BKT} \simeq 1.12$
(infinite-volume reference)
$\varepsilon = \pi, \quad S = -\kappa \sum_{\ell} \cos \theta_\ell$
L = 4
L = 8
L = 16
L = 32
10,000 sweeps per point • discard first 500 • 1σ block bootstrap